# Mahmoud Alyosify

## MLOps & Cloud AI Engineer

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## SUMMARY

MLOps and cloud AI engineer who takes trained models to **reliable, affordable services**. At Injaz Tech, wrote the Docker setup, a GitHub Actions workflow with 136 tests and an evaluation gate, and the health checks and metrics for a RAG service. Deployed a fine-tuned LLM on AWS, exported vision models to ONNX, and ran the master's thesis as reproducible benchmarks on an NVIDIA A40. MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30); three AWS Academy credentials.

## SKILLS

**Cloud & DevOps:** AWS, Docker, GitHub Actions, CI/CD, Linux, Bash
**Serving:** FastAPI, ONNX, FAISS, streaming APIs, caching, rate limiting
**Reliability:** automated testing (pytest), evaluation gates, monitoring (Prometheus), retries and circuit breakers
**Reproducibility:** pinned environments and models, deterministic runs, versioned data and results
**ML & data:** PyTorch, scikit-learn, Hugging Face, LangChain, SQL, PySpark, Pandas

## EXPERIENCE

**AI Engineer Intern** — Oct 2026 – Present
Injaz Tech (Injaz Group, formerly Fingerprint Consultancy) — Maadi, Cairo

- Wrote the **Docker** setup and a **GitHub Actions** workflow (linting, 136 tests, evaluation gate); added health checks, Prometheus metrics, retries and a circuit breaker.
- Built the **FastAPI** service: typed request and response models (Pydantic), streaming answers, session memory, health checks and rate limiting.
- Raised top-1 retrieval accuracy from **75% to 94%** and cut wrong refusals from 5 in 20 to 1 in 20.

**Systems Engineer (Military Service)** — Feb 2024 – Mar 2025
National Company for Roads Building & Development — Cairo

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles** across 70%+ of national roads.
- Debugged live transportation software and supported legacy applications to keep downtime to a minimum.

**Part-Time Machine Learning Instructor** — Jul 2025 – Present
National Telecommunication Institute (NTI) — Egypt

- Teach **90-hour machine learning bootcamps** and AI for Business modules in NTI's Summer Internships Program.

**AI in Cybersecurity Intern** — Jun 2025 – Jul 2025
e& Egypt (on-the-job training via NTI) — Cairo

- Applied AI tools to threat intelligence, digital forensics and enterprise security analysis.

**Online Instructor** — 2020 – Present
Udemy & YouTube (Arabic courses) — Remote

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

## PROJECTS

**Horus-OSINT — fine-tuned LLM on AWS** — 2026
*Python, Llama-3-8B, QLoRA, PySpark, AWS (EC2, S3, EMR), Terraform*

- Fine-tuned **Meta-Llama-3-8B** on **159,826** global event and terrorism records (GTD, GDELT) and deployed it on **AWS** as an assistant that answers questions in plain language.

**MinimalLM — Master's thesis, mentored by Deloitte's AI team** — 2026
*Python, PyTorch, Hugging Face Transformers, NVIDIA A40*

- Built to be reproduced: pinned models and GPU, greedy decoding, 56 automated tests, and a notebook that recomputes every number from the stored answers.
- Cost per token stayed within 0.8% of the original, so the time saved is real; GPU time in fixed batches fell 5–55%. Measured on an NVIDIA A40.

**SimCLR-Vision-SSL — self-supervised image model** 2026
*PyTorch, SimCLR, SupCon, ONNX, FAISS*

- Exported the model to **ONNX** and served image search through a **FAISS** index, ready to deploy.

**RL-Cloud-Autoscaler — deep RL for cloud scaling** 2026
*Python, Gymnasium, PPO, DQN*

- Built a cloud simulation (Gymnasium) and trained **PPO** and **DQN** agents to balance cost and latency, compared with standard threshold-based autoscaling.

**AudioShield — deepfake audio detection**
*Python, CNN, Streamlit*

- Trained a CNN on spectrograms to tell real speech from synthetic speech; served it in a Streamlit app with live predictions and PDF reports.

Also: HORUS Sentinel (multi-agent OSINT) · SCRAP (satellite collision risk, 48+ hours ahead) · Automated Multimodal Agent (PDF to narrated slides).

## PUBLICATIONS

**Say Less, Mean Better: Training-Free Verbosity Control for Language Models**
*M. Alyosify, M. Imbabi, E. Elkhamisi, R. Ferdousi · arXiv:2611.04107 (cs.CL) · November 2026*

**Beyond Truncation? Entailment-Licensed Elision and the Limits of Semantic Preservation in LLM Decoding**
*M. Alyosify, M. Imbabi, E. Elkhamisi · arXiv:2611.08975 (cs.CL) · November 2026*

## EDUCATION

**MSc in Artificial Intelligence** Sep 2025 – Oct 2026
Queen's University Kingston, Canada

- GPA **4.30 / 4.30** · Completed October 2026 · **Digilians Presidential Scholarship** (competitive national AI fellowship)
- Coursework: Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines

**BSc in Computer & Information Science (Bioinformatics)** Sep 2019 – Jul 2023
Assiut University Assiut, Egypt

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+
- Graduation project: Vitalism Solution, vital signs from ordinary video; **Microsoft Imagine Cup 2023 semi-finalist**

## CERTIFICATIONS

- AWS Certified Cloud Practitioner (Amazon Web Services)
- AWS Academy: Generative AI Foundations, Data Engineering, ML for NLP, ML Foundations
- IBM AI Developer Professional Certificate (IBM, 2026)
- Deep Learning Specialization (DeepLearning.AI, 2026)
- Machine Learning Specialization (DeepLearning.AI & Stanford, 2026)

## AWARDS & LEADERSHIP

- 1st Place, Smart Cities Hackathon 2022 (77 university teams)
- Semi-Finalist, Microsoft Imagine Cup 2023
- Finalist, ISEIC 2023 (international research competition)

## LANGUAGES

Arabic (native) · English (professional working proficiency)